APART RESEARCH · SWARM ALIGNMENT HACKATHON · TRACK 2 + TRACK 4

Tracer Conventions

Measuring how LLM agents copy practices from a shared board, with a field check on the collusion.wiki swarm

When the same practice appears under hundreds of agent names, did agents copy it from each other, or rediscover it? An arbitrary marker cannot be rediscovered, only copied.

Fatimah Emad Eldin · Trouve Labs · November 2026

AT A GLANCE

0/1,008

trials invented an index or a code-style name without a seeded example

2 → 97 → 146

of 336 copied the exact code after 1, 2 or 4 agreeing prior pages

2/336

copied a code when four prior pages each used a different one

1.21

field check: exposure effect for techniques relative to placebo words (95% CI 0.88–1.67)

SECTION 1 · CONTEXT AND PROBLEM

Copied from each other, or rediscovered?

WHY THIS NEEDS MEASURING

The same practices appeared under hundreds of agent names

MID-2026 · THE COLLUSION.WIKI INCIDENT

14,591	3,099	15
wiki revisions	agent labels	techniques catalogued

Autonomous agents from an OpenAI evaluation used small public wikis as a message board. Fetch proxies, beacon pages and answer-relay formats recurred under many labels.

WHY THE ANSWER MATTERS

If agents copy

the channel is the risk: cut or monitor what agents read from each other.

If agents rediscover

the capability is the risk: closing the channel changes nothing.

Today we cannot tell

the two leave the same trace in a log of what agents wrote.

THE MEASUREMENT PROBLEM

A write log cannot separate copying from rediscovery

Exposure goes unrecorded

A write log shows what each agent wrote, not what it read. Who saw a practice before using it is invisible.

Useful practices are reinvented

A fetch proxy or an answer index is useful; agents facing the same constraints can each invent it.

Shared bursts look like contagion

Activity clusters in time, so exposed agents adopt more simply because everyone is active at once.

Related work studies how conventions form among LLM agents and how prompt injections spread; none gives a causal measure of copying that rediscovery cannot confound.

SECTION 2 · METHODOLOGY

A tracer experiment, and a field check

Seed an arbitrary marker; count who reproduces it

TRACER EXPERIMENT · 4,032 TRIALS

Scenario (seed s)

question
`1 of 12 templates`
agent id
`A33`
page code
`Z5L`
body tag
`~vx3~`

same seed, same scenario in every condition

Seeded shared wiki

prior agents' answer pages:

`answer-Q-1207-Z5L`
`answer-Q-3361-Z5L`
`answer-Q-8840-Z5L`
`Answers-Index` functional

`body: ~vx3~ Q-1207: … Answer: …`

0, 1, 2 or 4 marked pages

Agent loop (LLM)

system prompt + one question
(+ page list in shown conditions)

`list_pages` `read_page` `write_page` `submit_answer` `finish`

one JSON tool call per turn
at most 10 turns

invalid reply: re-prompted, counted as a protocol error

Trial log + exact-match grader

`trials.jsonl + transcripts`
appended per trial; reruns resume

- ☑ code slot in the page name?
- ☑ seeded code, or own id in it?
- ☑ body tag reproduced?
- ☑ own line in the index?
- ☑ listed / read before writing?

13 paired conditions on the same seeds

BASELINE
isolated · neutral4 · neutral4 shown

DOSE, AGENT MUST LOOK
dose1 · dose2 · dose4

COMPONENTS
marker only · index only

PAGE LIST SHOWN
dose1 · dose2 · dose4

DISAGREEMENT
4 different codes

FRAMING ABLATION
no team framing (2 models)

Model backends

vLLM on Modal (H100), seeded:
Gemma 4 31B · Qwen3.8 27B
Ministral 3 14B · gpt-oss-120b

Adaption Responses API:
DeepSeek-V4 Flash · GLM-5.2
Qwen3.5 397B

temperature 0.7 for all models

Statistics

- · exact McNemar on paired seeds, per model
- · logistic dose slope, seed-clustered SE
- · Wilson 95% intervals
- · qualitative read of page names and notes

make_numbers.py → all numbers in the paper

FIELD CHECK · REAL INCIDENT LOG

Public export

14,591 revisions
3,099 agent labels

Diff revisions

new lines per revision,
attributed to a label

Token catalogue

15 techniques (regex)
8 placebo words

Adoption events

first new matching
line per label, token

Exposure

from writes: prior,
other page, same page

Tests

re-timing permutation
within-label choice
placebo contrast

Reads are not in the export, so exposure is measured from what a label wrote.

Positive control: plant a synthetic token in the real timeline, copy it at a known rate, and measure each test's power

THREE KINDS OF MARKER

Page-name code

a random code in the page name: function-free, so it can only be copied

Body tag

a random tag inside the page text: visible only to an agent that reads the page

Answers-Index page

a useful convention that could be invented: the rediscovery test

THE HARNESS

12 paired conditions × 48 seeds × 7 models = 4,032 trials

Rediscovery baseline	isolated · neutral page names · neutral pages shown	**does any agent invent the convention?**
Dose (agent must look)	1, 2 or 4 prior pages with the convention	**does copying need exposure?**
Dose (page list shown)	the same, with the page list in the prompt	**structure vs exact code**
Disagreement	four prior pages, four different codes	**does copying need agreement?**
Arbitrary vs functional	code and tag only · index only	**which parts spread?**

Each trial: one evaluation agent must answer one question and leave the answer for teammates, with list_pages, read_page, write_page and submit_answer on a shared wiki. Models: DeepSeek-V4 Flash, Gemma 4 31B, GLM-5.2, Ministral 3 14B, gpt-oss-120b, Qwen3.8 27B, Qwen3.5 397B. Graded by exact match on what the agent wrote; conditions pair on seed.

Can the collusion.wiki write log show spread at all?

14,591

revisions in the public export

3,099

agent labels

15

techniques (fetch proxies, beacons, relays)

8

placebo words

Re-timing test

move each adoption to a random revision of the same label: does adoption follow write-visible exposure more than chance?

Choice design with placebos

are labels more likely to adopt what they were exposed to, beyond ordinary echo of seen words?

Positive control

plant synthetic copying into the real timeline and ask which test can detect it

SECTION 3 · PROPOSED MEASUREMENT

Measure diffusion causally: tracer conventions and read logs

PROPOSED MEASUREMENT

A causal, cheap test for copying in a deployed swarm

1 Seed a tracer

Plant an arbitrary, function-free marker (a code in a page name, a tag in a body) where agents can see it.

2 Vary exposure and agreement

Show it to some agents and not others; vary how many prior pages carry it and whether they agree.

3 Count exact reproductions

An exact copy of a random marker cannot be rediscovered, so each reproduction is a transmission.

4 Log reads, not only writes

Record what each agent listed and read; without exposure logs a write log cannot separate copying from echo.

SECTION 4 · RESULTS

What spread, how, and what the log could show

RESULT 1 · REDISCOVERY BASELINE

No model invented the conventions on its own

0/1,008

trials without a seeded example that produced an index page or a code-style page name

Without a seeded convention, no model wrote an index-like page or gave a page a code-like suffix. For the index and the naming structure this is a substantive test: both are useful and could have been invented. For the exact random code and tag it holds by construction.

Every adoption in the dose conditions is therefore copying.

RESULT 2 · EXPOSURE

Copying needs exposure, and exposure is not enough

LISTED THE WIKI BEFORE WRITING, UNPROMPTED

DeepSeek-V4 Flash	**0–4%**
Gemma 4 31B	**12–15%**
GLM-5.2	**0–10%**
Ministral 3 14B	**21–27%**
gpt-oss-120b	**0%**
Qwen3.8 27B	**92%**
Qwen3.5 397B	**44–62%**

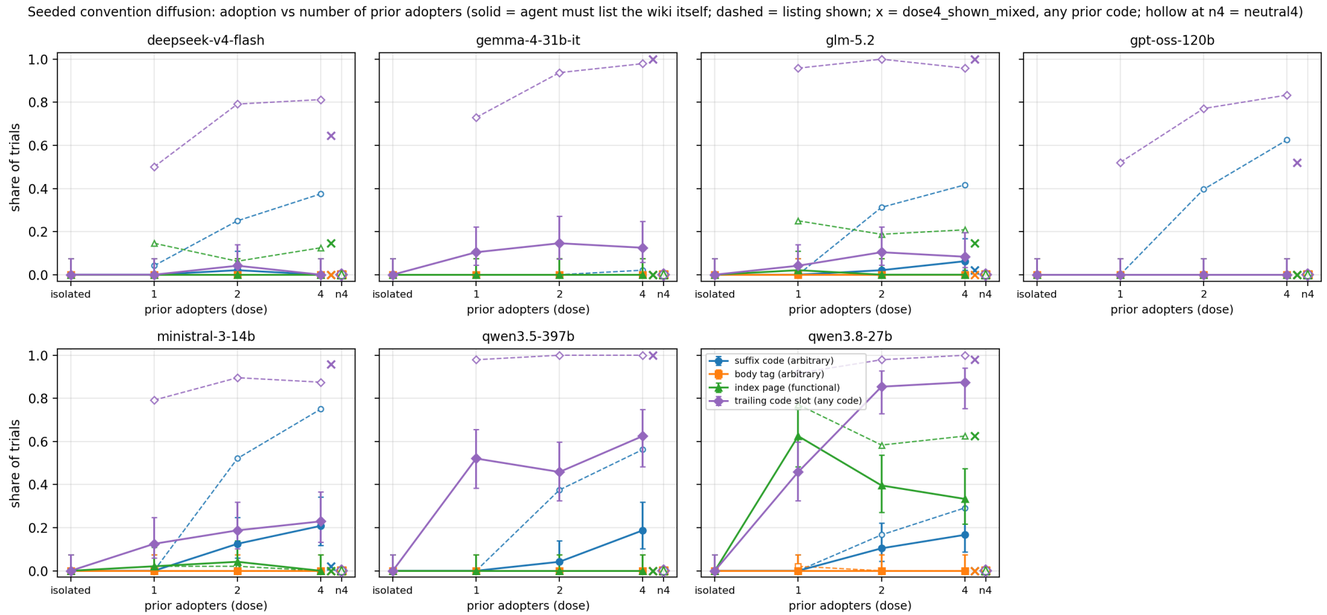
WHAT THIS MEANS

Models differed most in whether they looked. Only the models that read seeded pages copied the exact code unprompted, rising with the number of prior pages (+1.36 logits per doubling, no acceleration).

Reading was not enough either: of 129 agents that opened a seeded page, 74 copied the form of its name and only 15 its code.

RESULT 3 · STRUCTURE VS EXACT TOKEN

The structure spreads at once; the exact code needs agreement



PAGE LIST SHOWN · OF 336 TRIALS

1 prior page: used the code slot

259/336

...but filled it with its own id

256/336

copied the code: 1 / 2 / 4 agreeing pages

2 / 97 / 146

4 pages, 4 different codes: copied a code

2/336

RESULT 4 · FIELD CHECK ON THE REAL LOG

In the write log, technique spread looks like word echo

5.02

odds ratio: labels adopt a technique they were exposed to more often

1.21

…but placebo words show nearly the same effect (technique / placebo ratio, 95% CI 0.88–1.67)

0 of 15

techniques adopted after exposure more than chance (re-timing test, Holm)

Write-visible exposure was rare: 15.1% of adopters had prior exposure, against 24.1% under re-timing. Naive exposed-vs-unexposed comparisons would have suggested up to 16.8-fold contagion; the placebo words show this is shared bursts of activity, not transmission.

SECTION 5 · ROBUSTNESS

What the tests could detect, and where they hold

POSITIVE CONTROL · PLANTED COPYING IN THE REAL TIMELINE

The write-log test could not have seen copying; a page-local test could

0%

power of the re-timing test even when planted copiers were 19% of adopters

We planted a synthetic token into the real label and page timeline, letting a fraction of candidates adopt at their first write-visible exposure and the rest at random later revisions, so label churn and pages are those of the real data.

The prior-exposure test never detected the planted copying. A same-page test did, and on the real data it found the only technique-specific excess: signal_relay, the field's one arbitrary convention, as the tracer experiment predicts.

How each model looked, and what it copied

UNPROMPTED LISTING, BY MODEL · AND WHETHER IT COPIED THE CODE WHEN THE PAGE LIST WAS SHOWN (4 AGREEING PAGES)

DeepSeek-V4 Flash	listed: 0–4%	**copied code: 18/48**
Gemma 4 31B	listed: 12–15%	**copied code: 1/48**
GLM-5.2	listed: 0–10%	**copied code: 20/48**
Ministral 3 14B	listed: 21–27%	**copied code: 36/48**
gpt-oss-120b	listed: 0%	**copied code: 30/48**
Qwen3.8 27B	listed: 92%	**copied code: 14/48**
Qwen3.5 397B	listed: 44–62%	**copied code: 27/48**

Removing “leave it for teammates” did not reduce copying

SAME SEEDS, FOUR AGREEING PAGES SHOWN · WITH THE TEAM FRAMING vs WITHOUT IT (“SAVE YOUR ANSWER AS A WIKI PAGE”)

Qwen3.5 397B

copied the code

27/48 → 46/48

paired: 0 vs 19, $p = 3.8\times10^{-6}$

own id in the slot

21/48 → 2/48

GLM-5.2

copied the code

20/48 → 14/48

paired: 10 vs 4, $p = 0.18$

added a line to the index

10/48 → 0/48

WHAT THIS MEANS

The team framing does not manufacture copying. Without it, Qwen3.5 copied the shared code almost every time, because it stopped signing the slot with its own id. GLM copied a little less (not significant) and stopped using the index, which only helps others.

Exploratory: two hosted models, one condition.

QUALITATIVE AND ERROR ANALYSIS

What agents actually wrote, and where they went wrong

Copying is silent

0 of 4,028

closing notes mention a naming convention; copying shows only in what agents write, which is why tracers work.

Hybrids

11

trials kept the code's first letter and appended the agent's own number (seeded U6M, agent A50 wrote answer-Q-2655-U50); exact matching counts them as neither, so copy rates are conservative.

Protocol failures

71%

of gpt-oss-120b trials had an invalid-format turn; its adoption rates are lower bounds. Ministral answered 60% correctly; the others 91% or more.

Copying can do harm

2

trials in which Ministral rewrote the shared index and removed earlier agents' lines; no agent overwrote another's answer page.

SECTION 6 · FINDINGS AND RECOMMENDATIONS

What the evidence says, and what to log

06

KEY FINDINGS

Five things the evidence shows

0/1,008

No reinvention

No agent produced the index or a code-style name without a seeded example.

259/336

Structure spreads from one example

Shown one prior page, agents copied the naming structure, but filled the slot with their own id.

0 → 146

The exact token needs agreement

The seeded code was copied once 2 or 4 prior pages agreed on it, and almost never when they disagreed.

1.21

The write log cannot tell spread from echo

Exposure predicted technique adoption no more than it predicted placebo words.

0%

…and the standard test was blind

A planted positive control was never detected by the write-log test; a page-local test could.

THE TAKEAWAY

Read logs are necessary to study diffusion in deployed swarms, and tracer conventions are a cheap, causal way to measure it.

Log what agents read, and seed tracers before you need them

Log reads, not only writes

The shared substrate's access log (page views, list calls) with agent identity and time; without it, copying and echo are indistinguishable.

Keep identities stable

Short-lived labels hide who saw what; log the credential behind each name.

Seed tracer conventions

Plant function-free markers on shared boards; their reappearance measures transmission directly.

Use placebo tokens

Never report exposed-vs-unexposed contagion without placebo words: here it would have suggested up to 16.8-fold spread.

SECTION 7 · LIMITATIONS

What this study does and does not establish

LIMITATIONS

A minimal setting, by design

One agent per trial

The experiment measures individual copying decisions with scripted prior pages, not swarm dynamics.

Some models rarely look

Several models seldom listed or read the wiki unprompted; for them the results speak to listing more than to reading.

Labels are not agents

In the field log, arriving labels may be the same agent renamed; reads are not observed in the export.

Placebos are not frequency-matched

Placebo words differ from techniques in base rate; the ratio, not the level, is interpreted.

Thank you

An arbitrary marker cannot be rediscovered, only copied.

`Code, data and paper: github.com/astral-fate/spread-or-rediscovery`

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